Pub/Sub Options - Draft

Referrals and Appointments

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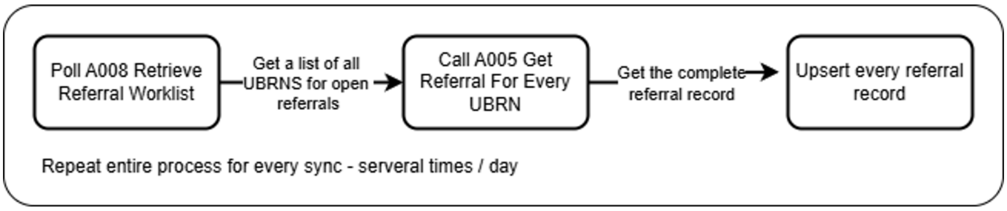
Background - The Problem We Would Like to Solve

Many of our integration partners need to maintain an up to date database of the referrals which are relevant to them. In order to obtain the referral records they are using the eRS external API. The API does not offer any endpoints which are well suited to this task. The typical strategy is to regularly poll the A008 Retrieve Referral Worklist end point in order to obtain a list of UBRNS. Then poll the A005 endpoint for each referral, each time an update is required. There is no mechanism to discover if the state of a referral has changed, other than to retrieve the complete referral details.

As a result of this, integrating systems have no choice but to large amounts of data out of eRS in order to keep their records in sync. The majority of this data is redundant. This puts a high load on the eRS system, while providing a poor integration experience for partners; many users experience long sync delays between eRS and local systems.

An additional challenge for integrators is that worklists as all associated with users, and are available via user restricted access only. For many sites syncing referrals requires someone to log in and press a button every time.



Summary: integrating systems must pull all their current referral data every time they sync in order to get any change or update.

1 Key Issues:

• Worklists not an ideal mapping for integration purposes - '*we don't care about your worklists, we have our own, we just want all the referrals*' - Nervecentre
• Large amount of data has to be transferred in order to detect changes
• Updates are slow, only a couple of times a day
• User action required to run sync process

2 What Are We Looking For In A Solution?

• Ability to get a list of all the referrals relevant to a site / organisation without depending on worklists
• Ability to easily tell if a referral has changed
• Ability to get immediate notifications of referrals that are relevant to a site of organisation which have changed
• Ability to perform these actions using application restricted access
• A relatively accessible upgrade path for integrators

With the goal of providing fast, reliable referral synchronisation between eRS and partner systems without the need for human intervention.

3 Three Stage Solution

3.1 Stage One - The Quick Fix

Include referral version id in worklist payload.

Already in progress, we are adding a version number to the referral details returned by the A008 Retrieve Referral Worklist endpoint.

This will remove the need for integrating systems to poll every referral that is currently relevant to them. Instead they call poll the worklist, check the version numbers against those they currently hold and only request details for the referrals which have changed.

Clearly this is not a complete fix, but it reduces the load on eRS considerably and we can get it done quickly.

What it fixes

- Ability to easily tell if a referral has changed
- A relatively accessible upgrade path for integrators

What it does not fix

- Ability to get a list of all the referrals relevant to a site / organisation - no improvement
- Ability to get immediate notifications of referrals that are relevant to a site of organisation which have changed
- Ability to perform these actions using application restricted access

3.2 Stage Two - The Foundation

New get referrals endpoint that is not tied to the worklist concept.

Create a new integration focused Get Referrals API endpoint. This would allow a list of referrals to be retrieved with filter criteria applied. Specifics TBA, the intention is for an integrating system to provide something like a provider ODS code or a referrer ODS code and get a list of referrals which are relevant to them. There may be an option to filter by referral status, date ranges, etc.

This would return a list of referral IDs along with their version numbers.

Ideally this would be BaRS compliant, based on Get referral/s for patient:
https://digital.nhs.uk/developer/api-catalogue/booking-and-referral-fhir/v1.3.0#get-/ServiceRequest

This provides a much more efficient polling mechanism than the worklist concept, and is better suited to application-restricted access.

** This stage is a pre-requisite for stage three **

What it fixes

- Ability to get a list of all the referrals relevant to a site / organisation
- Ability to easily tell if a referral has changed
- Ability to perform these actions using application restricted access

What it does not fix

- Ability to get immediate notifications of referrals that are relevant to a site of organisation which have changed

Discuss

- A relatively accessible upgrade path for integrators .. Or is it?
- Potentially enables other initiatives

3.3 Stage Three - The Final Fix

Pub/Sub is the standard solution for push notifications, rather than polling. The Multicast Notification Service provides the facilities needed to implement pub/sub in the context of eRS and it's wider partner environment.

https://digital.nhs.uk/developer/api-catalogue/multicast-notification-service

High Level Overview - What MNS Can Do

MNS allows us to easily distribute referral update notification events.

Messages contain a UBRN, version id & URL to retrieve the full referral detail.

Subscribers can choose to receive their notifications either through a MESH mailbox or via an AWS SQS queue. The service hides all the detail of dealing with MESH from us - we just publish our events by posting them to a REST endpoint.

Subscribers can choose the message types and filters they require when they subscribe, e.g. 'R4 Referral Updates for ODS code XXX'.

Event Message Content & FHIR Versions

MNS conceptually supports multiple versions of messages for the same event, which would allow for emitting notifications which refer to both STU3 & BaRS r4 end points.

Subscription Management Options

- Integrator On boards with MNS direct and manages subscriptions direct
- eRS provide subscription management APIs and manages subscriptions on the integrators behalf - recommended

Add note on lightweight mvp with manual management

The MNS team have expressed a preference for the latter option. It also simplifies the process for partners, removing the need for them to go through another onboarding process.

Reference Integration Pattern

- Subscribe by [ODS Code] - To get ongoing notifications
- Call Get Referrals (by [ODS Code]) - To get existing inflight referrals
- On receipt of change notification - check if the referral id is known - no create it, yes - check the version id and update if changed
- Do not depend on events for SLAs - Poll Get Referrals frequently enough to ensure SLAs are met [specific recommendation here]

Service Level Mismatch

eRS is a gold level service, while MNS is bronze. We can account for this with the 'Do not depend on events for SLAs' clause in the reference integration pattern.

What it fixes

- Ability to get a list of all the referrals relevant to a site / organisation
- Ability to easily tell if a referral has changed
- Ability to get immediate notifications of referrals that are relevant to a site of organisation which have changed
- Ability to perform these actions using application restricted access

What it does not fix

- N/A

Discuss

- A relatively accessible upgrade path for integrators